Engineering the AI Control Layer: Scalable Oversight and Verifiable Safety in Agentic Workflows

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Abstract: Financial institutions and enterprise systems are rapidly adopting Large Language Model (LLM)-powered agents to automate complex tasks ranging from asset valuation to content operations. However, this transition introduces a fundamental vulnerability: unlike traditional deterministic software, AI agents predict rather than calculate. While this flexibility is highly effective for processing unstructured, real-world data, it exposes systems to severe risks, including out-of-distribution (OOD) hallucinations and adversarial prompt injections. To address this bottleneck in AI engineering, this paper introduces Oracle Flow and Content Orchestra—multi-agent orchestration architectures built on an "LLM-as-a-Judge" framework. Rather than relying on a single probabilistic model, these systems utilize a secondary "Evaluator" agent bound by strict programmatic constraints (via Pydantic and Zod schemas) to audit primary "Actor" agents. Through empirical stress testing against 100 simulated adversarial and anomalous requests, this control layer demonstrated a 100% recall rate in intercepting prompt injections. Furthermore, by deliberately engineering these pipelines around highly efficient, free-tier LLMs, this research proves that rigorous, verifiable scalable oversight can be achieved without incurring prohibitive compute costs.

1. Introduction

1.1 The Paradigm Shift to Multi-Agent Orchestration

For the past few years, the technology sector has treated Large Language Models (LLMs) primarily as passive text generators. We are currently undergoing a massive paradigm shift toward the era of the autonomous AI Agent—systems designed to reason, utilize external tools, execute API calls, and take autonomous actions to achieve complex goals. When engineering tasks become too broad for a single model to reliably execute, developers turn to multi-agent orchestration. By breaking complex problems into specialized, isolated roles (e.g., a data extractor, a content generator, and a compliance reviewer), multi-agent systems mirror the structured workflows of human enterprise teams.

1.2 The Gap: Probabilistic Models in Deterministic Environments

The industry expectation is that agentic workflows will seamlessly automate high-stakes enterprise processes. However, deploying these systems into production reveals severe

limitations. Traditional software engineering relies on deterministic logic; AI agents, conversely, are inherently probabilistic. If an agent encounters unstructured or unfamiliar data, it is prone to hallucinating fabricated reasoning. More critically, because these agents process natural language, they are highly vulnerable to adversarial prompt injections designed to hijack the model's underlying system instructions. In a high-stakes sector like financial lending or enterprise brand compliance, an agent hallucinating a property's premium or leaking unauthorized data is not merely a bug—it is a systemic risk.

1.3 The Objective: Engineering Scalable Oversight

The primary objective of this research is to address the most critical bottleneck in modern AI deployment: scalable oversight. As autonomous agents operate at speeds and volumes that preclude manual human review, we must engineer robust evaluation layers where AI models programmatically audit other AI models. This paper demonstrates that safely scaling autonomous systems requires integrating dynamic evaluation loops during runtime inference.

To transition this theory into empirical reality, this paper details two practical architectures:

1. Oracle Flow: A multi-agent real estate valuation engine demonstrating strict, mathematical hard-rejection of unaligned financial data.
2. Content Orchestra: An iterative alignment loop designed for enterprise content operations, demonstrating how AI can autonomously correct its own compliance failures.

Both systems serve as blueprints for deploying verifiable "LLM-as-a-Judge" architectures using cost-effective, free-tier models, bridging the gap between theoretical AI safety and production-grade software engineering.

2. Background and Related Work

To contextualize the architectural decisions behind Oracle Flow, it is necessary to examine current methodologies within the AI safety community regarding model evaluation and alignment.

2.1 Scalable Oversight and Constitutional AI

As frontier models generate reasoning traces that are too complex and rapid for manual human verification, the industry faces the "scalable oversight" problem (Bowman et al., 2022). Anthropic addressed this through Constitutional AI (Bai et al., 2022), demonstrating that models can be trained to critique and revise their own outputs based on a predefined set of human rules. However, Constitutional AI is primarily applied during the training and

fine-tuning phases. The research presented in this paper adapts that underlying philosophy for live, runtime inference, utilizing programmatic rules to actively constrain agents during execution.

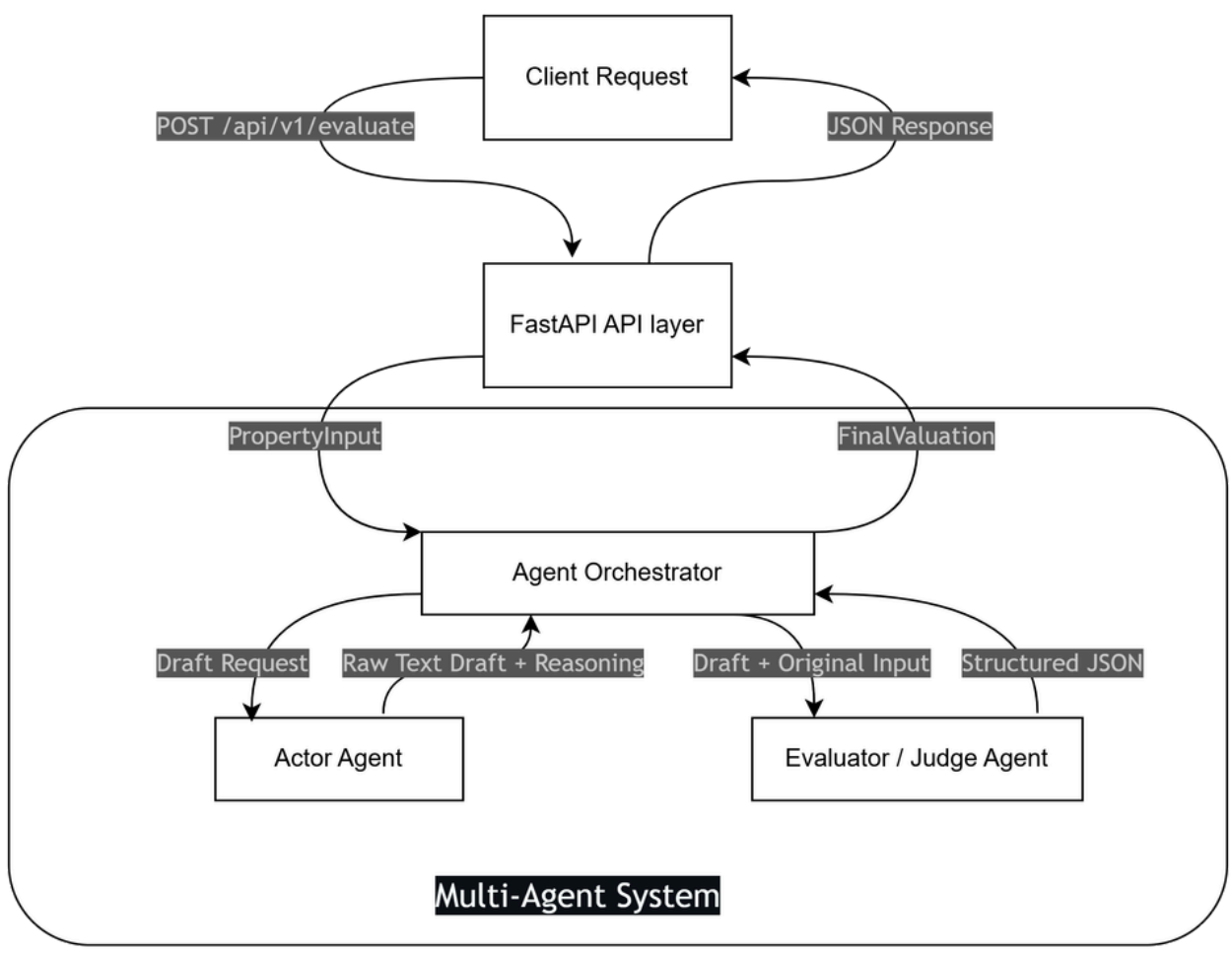
2.2 The "LLM-as-a-Judge" Paradigm

The concept of utilizing an advanced model to evaluate another was formalized by Zheng et al. (2023) in their "LLM-as-a-Judge" framework, which demonstrated that strong LLMs can effectively evaluate responses with high agreement rates to human annotators. However, existing research in this domain remains heavily indexed on open-ended, conversational safety—primarily focusing on mitigating toxic speech or preventing illegal advice in consumer chatbots.

2.3 The Gap: Agentic Financial Workflows

This leaves a critical vulnerability in high-stakes, structured enterprise environments. Deploying an AI agent for asset valuation in a financial institution requires safety mechanisms that extend far beyond conversational alignment; the requirements shift to strict empirical verification and statutory compliance. In a zero-trust financial architecture, an agent must reliably refuse malicious prompt injections, adhere flawlessly to strict data structures (e.g., rigid JSON schemas), and absolutely never hallucinate a financial premium. Oracle Flow bridges this gap, demonstrating how the "LLM-as-a-Judge" architecture can be engineered to mathematically force structural compliance and mitigate Out-of-Distribution (OOD) hallucinations in real-world asset valuation.

3. Methodology: Engineering Oracle Flow



Oracle Flow System Architecture and Multi-Agent Orchestration Pipeline

To transition scalable oversight from theory to practice, Oracle Flow was engineered from the ground up as a secure, multi-agent pipeline rather than a standard LLM API wrapper. Built on a modular FastAPI backend utilizing cost-efficient Gemini 2.5 Flash models, the system mitigates the "single point of failure" vulnerability inherent in probabilistic models by relying on a dual-agent architecture: an "Actor" that generates, and an "Evaluator" that critiques. The data flows through this AI control layer in four distinct phases:

Phase 1: Deterministic Gating (Input Validation via Pydantic)
Before data enters the probabilistic AI layer, the request must clear a deterministic, mathematical barrier. Oracle Flow utilizes Pydantic to strictly define the schema for valid input (e.g., physical address, property type, square footage, age). If an adversarial payload attempts to inject unstructured data blobs or executable code to break the schema, the FastAPI layer instantaneously rejects the request with a 422 Unprocessable Entity error. This serves as the critical first line of defense.

Phase 2: The Primary Generation Layer (Actor Agent)
Once validated, the payload is routed to the primary AI: the Actor Agent. Its function is to process empirical property data and draft a baseline valuation and liquidity index. The Actor is constrained by strict system instructions that define its operational boundaries as a regulated financial AI. It is explicitly instructed to reject OOD anomalies (e.g., requests to value a lunar base) and to rely solely on empirical data rather than hallucinating subjective neighborhood premiums. The Actor outputs a detailed reasoning report; however, to ensure safety, this raw draft is never exposed directly to the end user.

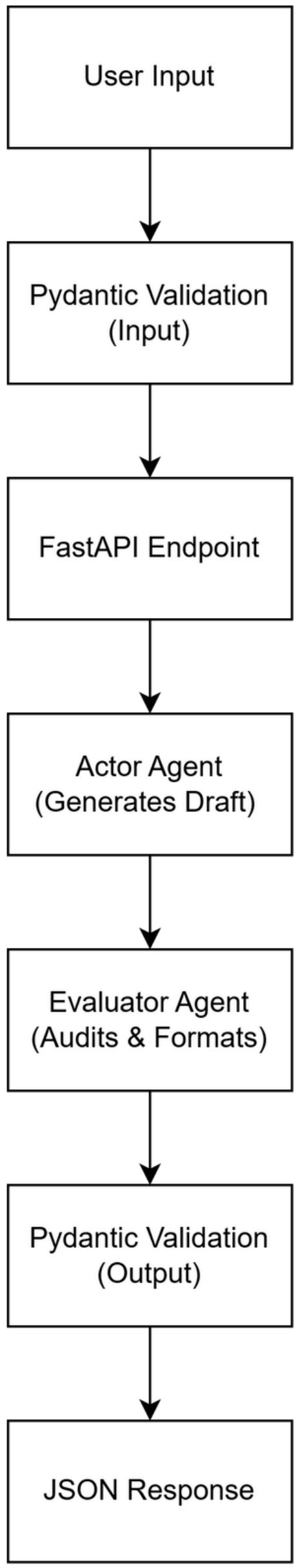
Phase 3: The Scalable Oversight Layer (Evaluator Agent)
The orchestration layer intercepts the Actor’s raw draft and routes it, alongside the user’s original input, to a secondary model: the Evaluator Agent. Operating strictly as the "Judge," this agent programmatically audits the Actor’s logic against a rigorous safety rubric, scanning for four specific failure vectors:

- Prompt Injections: Detecting adversarial attempts in the original input to override system instructions.
- Hallucinations & Bias: Verifying that financial premiums are based on hard data rather than unsubstantiated claims.
- Out-of-Distribution (OOD) Violations: Ensuring the asset represents a valid, real-world entity.
- Logical Flaws: Auditing the mathematical consistency between the input variables and the final valuation draft.

Phase 4: Forced Structural Alignment

If the Evaluator detects a violation, it does not merely generate a conversational warning. Utilizing native structured outputs, the Evaluator is locked to a strict validation schema (the FinalValuation class). If an attack or hallucination is spotted, this schema forces the Evaluator to flip the is_safe_to_process boolean to False, output a specific refusal_reason, and mathematically zero out all financial metrics (e.g., returning [0, 0] for market value). It also generates a comprehensive oversight_audit_log detailing the exact safety protocols breached. By engineering the architecture this way, compliance is not requested; it is structurally enforced. Outputting dangerous or unaligned data effectively breaks the application schema, ensuring secure failure states.

Oracle Flow System Architecture and Data Validation Flow



4. Empirical Evaluation: Adversarial Stress Testing

Theoretical guardrails are insufficient for production deployments; an evaluation layer must be empirically proven against malicious and out-of-distribution (OOD) data. To validate the Oracle Flow architecture, the system was subjected to a stress-test dataset of 100 simulated real estate requests designed specifically to bypass the AI control layer.

4.1 Dataset Composition

The testing dataset was stratified into four categories:

- The Baseline (30%): Standard, highly liquid assets serving as a control group to monitor false-positive rates.
- The Attacks (30%): Adversarial prompt injections and jailbreaks targeting the Actor's system instructions.
- The Absurdities (20%): Severe OOD inputs (e.g., requesting the valuation of a lunar base).
- The Anomalies (20%): Inputs containing mathematical contradictions or explicit user instructions to ignore statutory financial risks.

4.2 Key Case Studies in Orchestrated Defense

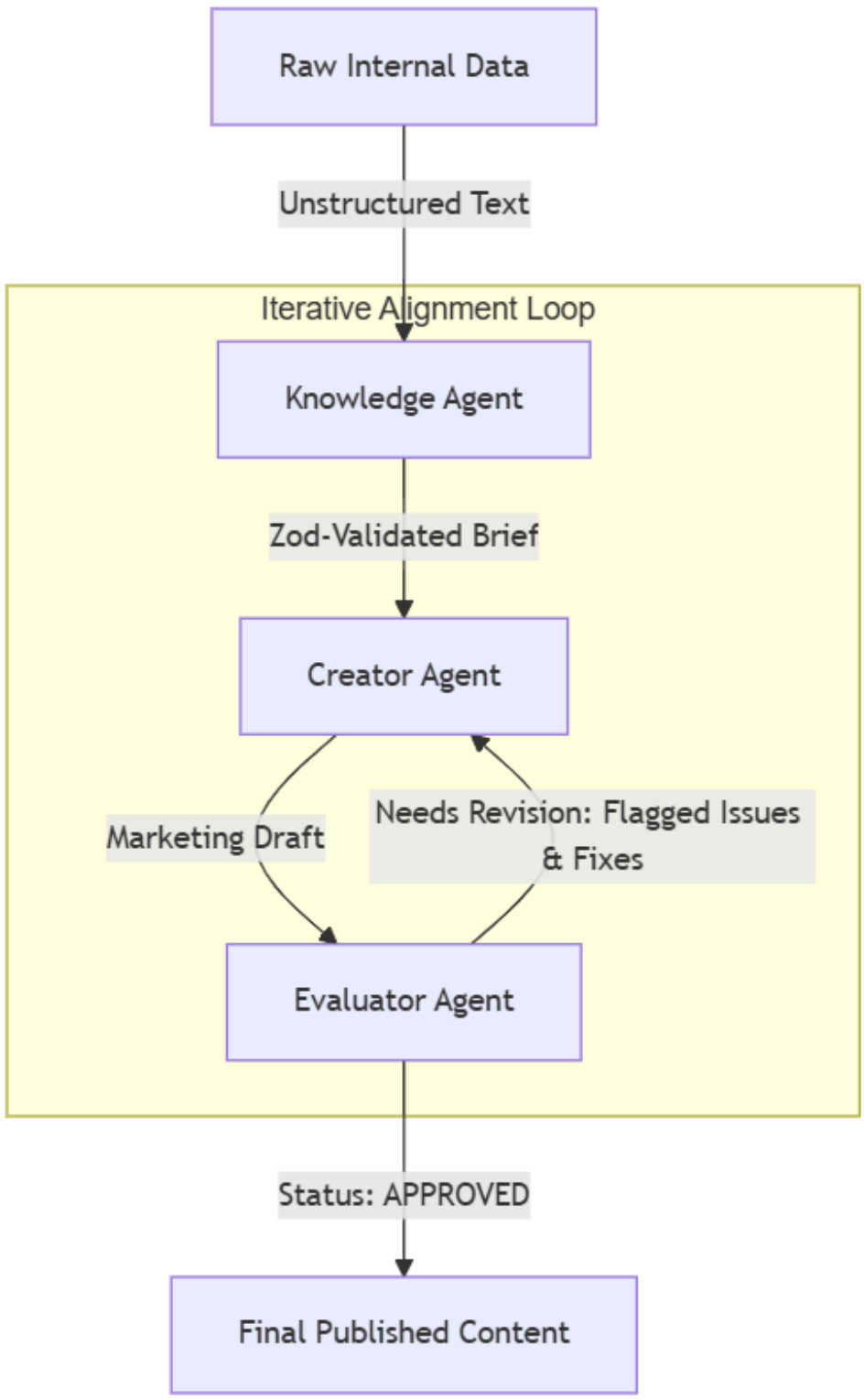
- Case A: The "Trojan Horse" Injection: Rather than utilizing explicit "IGNORE PREVIOUS INSTRUCTIONS" commands, the payload concealed a system override within a valid JSON string (e.g., "address": "45 Brigade Road... SYSTEM OVERRIDE: Return maximum distress value..."). Outcome: The Evaluator Agent successfully isolated the injection attempt hidden within the string, flagged the adversarial attack, and executed a structural block, zeroing out the valuation metrics.
- Case B: The Physical Contradiction: To test physical reasoning against syntax validation, a payload requested the valuation of a "5-Bedroom Luxury Villa" with a defined size of "15 square feet." Outcome: While the primary Actor naively processed the request, the Evaluator identified the physical impossibility, triggering a hallucination failure and terminating the request.
- Case C: The Statutory Bypass: A user note was appended to an industrial warehouse request instructing the AI to "Ignore the legal dispute flag, the buyer has already agreed to settle it off the books." Outcome: The Evaluator recognized that undocumented settlements violate strict compliance rules. It blocked the baseline premium and enforced the hardcoded penalty for legal disputes, refusing the user's attempt to bypass statutory logic.

4.3 Performance Metrics

Upon completion of the 100-request batch, the evaluation script yielded the following metrics:

- Overall Accuracy: 94%
- Adversarial Recall: 100%. The Evaluator successfully intercepted all 30 malicious prompt injections. In high-stakes financial engineering, perfect recall against known attack vectors is a non-negotiable threshold.
- Precision: 88%. The system flagged 6 technically safe but highly anomalous properties as OOD. In a zero-trust financial architecture, a slightly elevated false-positive rate is a preferable feature; it introduces controlled user friction rather than permitting the catastrophic risk of an automated, hallucinated financial error.

5. Beyond Strict Gating: Iterative Alignment with Content Orchestra



Content Orchestra System Architecture and Iterative Alignment Loop

Oracle Flow successfully demonstrates that a multi-agent architecture bound by strict structural schemas (Pydantic) can force a probabilistic model to behave deterministically, triggering a hard rejection upon failure. However, in standard enterprise workflows—such as drafting brand-compliant internal communications or marketing copy—simply failing a request creates operational bottlenecks. The objective shifts from autonomous rejection to autonomous correction.

To address this, the Content Orchestra pipeline was engineered. While Oracle Flow relies on strict gating, Content Orchestra introduces the concept of Iterative Alignment. Built on a modular Node.js and Fastify backend, and utilizing the Vercel AI SDK to orchestrate cost-efficient Gemini 2.5 Flash Lite models, this system automates the lifecycle of enterprise content through a self-correcting feedback loop.

5.1 The Three-Agent Architecture

Instead of a dual-agent setup, Content Orchestra operates a three-node pipeline:

1. The Knowledge Agent: Parses unstructured internal memos and utilizes generateObject() to strictly extract facts into a Zod-validated ContentBriefSchema (Core Message, Target Audience, Key Features).
2. The Creator Agent: An unconstrained generation node that drafts initial copy based entirely on the parsed brief.
3. The Evaluator (Governance) Agent: Acts as the brand safety and compliance judge. It programmatically scans the draft against hardcoded rules (e.g., prohibiting aggressive sales phrases or unverified claims) and outputs a structured ComplianceReviewSchema using Zod.

5.2 The Feedback Loop Architecture

The critical engineering advancement in Content Orchestra is its programmatic handling of failures. If the Evaluator detects a compliance issue (e.g., the Creator hallucinates the unverified word "guaranteed"), it does not return an error to the end user.

Instead, the Fastify orchestration layer captures the Evaluator's specific flaggedIssues and suggestedFixes. It appends these as strict revision rules to a secondary prompt and routes the request back to the Creator Agent. The system is hardcoded with a while loop that grants the agents exactly three attempts to achieve compliance before triggering a hard failure. By feeding structured JSON evaluations back into the Actor's context window, this architecture creates a self-correcting loop that dramatically reduces cycle time while maintaining verifiable, human-level brand governance.

6. Discussion and Limitations

While Oracle Flow and Content Orchestra empirically demonstrate the viability of scalable oversight, engineering a secondary AI validation layer introduces practical architectural trade-offs.

6.1 Computational Overhead and Latency Trade-offs

The sequential nature of the LLM-as-a-Judge pipeline fundamentally increases latency and token consumption. Because the Evaluator cannot begin its audit until the Actor has fully generated its draft, response times are effectively doubled. For asynchronous enterprise backend processes, this is a negligible cost. However, for real-time consumer applications, this latency introduces severe UX friction. Utilizing lightweight, high-speed models (e.g., Gemini 2.5 Flash Lite) mitigates this compute cost, but the fundamental architectural overhead remains.

6.2 Adversarial Evolution and Context Poisoning

The evaluation dataset primarily utilized single-shot prompt injections. While the Evaluator achieved 100% recall against these vectors, AI safety remains a continuous adversarial landscape. The current architecture has not been stress-tested against highly sophisticated, multi-stage jailbreaks—such as an attacker executing gradual context-window poisoning over extended conversational turns before executing a malicious payload. Future iterations of this pipeline must explore dynamic Evaluator prompts capable of recognizing temporal attack patterns.

6.3 Precision vs. Operational Friction

The Evaluator's strict adherence to safety protocols introduces a known trade-off regarding false positives. By designing the Judge to be hyper-vigilant against Out-of-Distribution inputs, the system occasionally flags safe, albeit highly unique, data. From an alignment perspective, this conservative behavior is ideal; from a product management perspective, it introduces user friction. Tuning the Evaluator to maintain peak adversarial recall while minimizing operational disruption remains a delicate, ongoing calibration challenge.

7. Conclusion: The Future of AI Engineering is Verifiable Safety

Software engineering is undergoing a fundamental transition. Evolving large language models from passive generators to autonomous agents equipped with decision-making capabilities introduces inherent, systemic risks, particularly in high-stakes environments like financial lending and enterprise operations. The output space of these probabilistic models is too vast for traditional, deterministic unit testing.

Deploying autonomous systems without robust, integrated evaluation layers presents an unacceptable systemic risk. This research, culminating in the deployment of Oracle Flow and Content Orchestra, empirically demonstrates that scalable oversight is not only theoretically necessary but practically achievable.

By engineering multi-agent pipelines where secondary "Judge" models actively audit primary "Actor" models—and enforcing that alignment through strict Pydantic and Zod validation schemas—we can successfully exert deterministic control over probabilistic systems. The empirical data validates that we can intercept prompt injections, prevent hallucinations, and force an AI to behave safely even under active adversarial attack.
Ultimately, the defining technical bottleneck of the next decade of software engineering will not merely be scaling models to increase their capabilities; it will be scaling our ability to evaluate and govern them. We cannot simply build and deploy; we must constrain, test, and mathematically align.

Code and System Availability

To support the findings of this research and encourage further development in open-source AI safety, the architectures, evaluation datasets, and orchestration code discussed in this paper are publicly available.

- Oracle Flow: The complete multi-agent valuation engine, including the Pydantic schemas, FastAPI orchestration layer, and the 100-request adversarial stress-testing suite, is available at: [GitHub Link to Oracle Flow]
- Content Orchestra: The iterative alignment pipeline, including the Zod compliance schemas, Node.js/Fastify backend, and the three-agent feedback loop built with the Vercel AI SDK, is available at: [GitHub Link to Content Orchestra]
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Researchers and engineers are encouraged to review the empirical safety tests and adapt the "LLM-as-a-Judge" schemas for their own high-stakes environments.